

BIOEQUIVALENCE OF SITAGLIPTIN, METFORMIN AND GLIMEPIRIDE TABLETS IN HEALTHY ADULT VOLUNTEERS: RESULTS FROM A RANDOMIZED, TWO-WAY CROSSOVER, OPEN-LABEL STUDY

Fixed dose combination of Sitagliptin + Glimepiride + Metformin – Rationale



101.3 Million People have diabetes in India with ~67% patients not reaching their HbA1c targets Average HbA1c is 8.9%

Many diabetic patients on Metformin + Sulphonylurea (SU) are unable to achieve control and require SU dose escalation

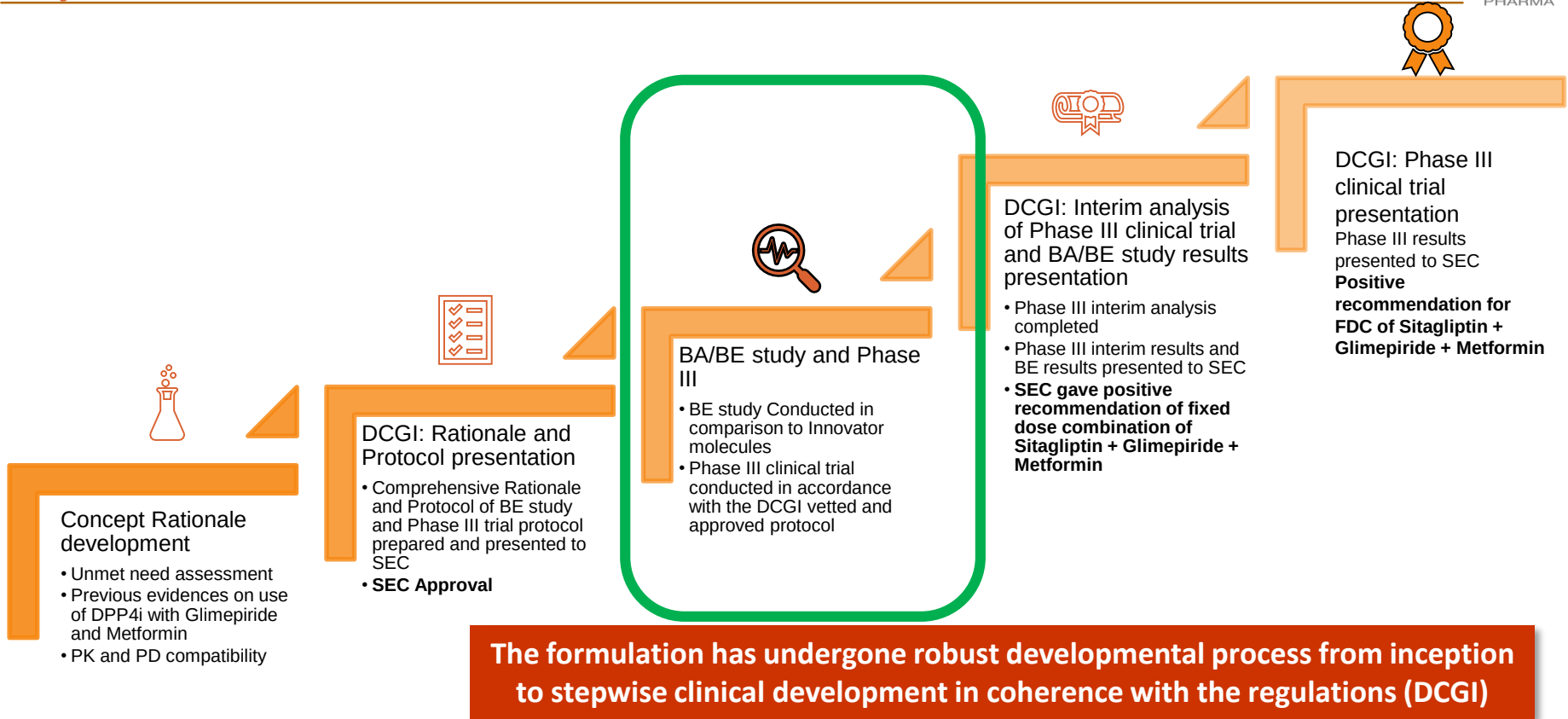
With over 30% patients taking 3-4 tablets/day, pill burden results in poor treatment adherence, resulting in inadequate glycemic control

Combining DPP4 inhibitors, SU with Metformin can help achieve glycemic goals with complimentary MOAs

- Multiple Pathophysiologic mechanisms addressed: Metformin ↓ Insulin Resistance, Glimepiride ↑ Insulin Secretion
- Addition of Sitagliptin: Complementary mechanism for glucose induced insulin secretion, Sitagliptin exhibits known synergism with Metformin through several mechanism
- β-cell function: Addition of Sitagliptin to glimepiride shown to preserve beta cell function (lower PI/I ratio in studies)

In this context we developed FDC of Sitagliptin + Glimepiride + Metformin formulation which has undergone robust developmental process.

Stages of Clinical Development of Sitagliptin + Glimepiride + Metformin Hydrochloride IR FDC



Study Design	Single dose two-way crossover bioequivalence study on Sitagliptin Phosphate, Metformin Hydrochloride and Glimepiride 50 mg/ 1000 mg/ 2 mg tablets in healthy, adult, human subjects under fed condition.
Study Objective	The objectives of this study are to assess the bioequivalence between Test product (T) and Reference product (R) and to monitor the safety of Test product (T) and Reference product (R) in healthy adult human subjects under fed condition.
Investigational Product Details	Reference (R): Reference Product (R1) immediately followed by Reference Product (R2) will be administered and will be coded as Reference product (R). Reference (R1): Janumet® (Sitagliptin phosphate & Metformin Hydrochloride tablets) 50mg/1000 mg manufactured by MSD Pharmaceuticals Pvt. Ltd., India Reference (R2): Amaryl® (Glimepiride Tablets I.P.) 2 mg manufactured by Sanofi India Limited Test (T): Sitagliptin Phosphate, Metformin Hydrochloride and Glimepiride tablets 50 mg / 1000 mg / 2 mg, manufactured by Sun Pharma Laboratories Limited (SPLL)

Dosing Schedule	A single oral dose of either Test (T) or Reference (R) products will be administered as follows in each period of the study.
Study Population	Forty Eight (48) healthy adult human subjects
No. of volunteers	The study was conducted with 48 subjects (41 completed the study)
Study Dosing Date	Period-I: 15-Jul-2021 and Period-II: 23-Jul-2021
Washout	08 days
Primary PK Parameters	C_{max} , AUC_{0-t} and AUC_{0-inf} for Sitagliptin, Metformin and Glimepiride.

Key

Inclusion Criteria

□ Subjects included;

- Were in the age range of 18-50 years.
- Had non-vegetarian diet habit.
- Had body mass index (BMI) ≥ 18.50 and < 30.00 kg/m².
- Had voluntarily given written informed consent to participate in this study.
- Subject was healthy as determined by medical history and physical examination performed within 28 days prior to the commencement of the study.
- Had hemoglobin level ≥ 12.0 g/dL (Deciliter) for males or ≥ 11.0 g/dL for females.
- Had clinically acceptable results for all the screening parameters and laboratory investigations.
- If subject was female and:
 - a) Of childbearing potential, was practicing or willing to practice an acceptable method of birth control for the duration of the study as judged by the investigator(s), such as condoms, foams, jellies, diaphragm, intrauterine device (IUD), or abstinence; or
 - b) Was post-menopausal (Post-menopausal status is defined as absence of menses for the past 12 months or hysterectomy with bilateral oophorectomy at least 6 months ago) or
 - c) Was surgically sterile (Hysterectomy, bilateral oophorectomy or bilateral tubal ligation at least 6 months ago).

Key Exclusion Criteria

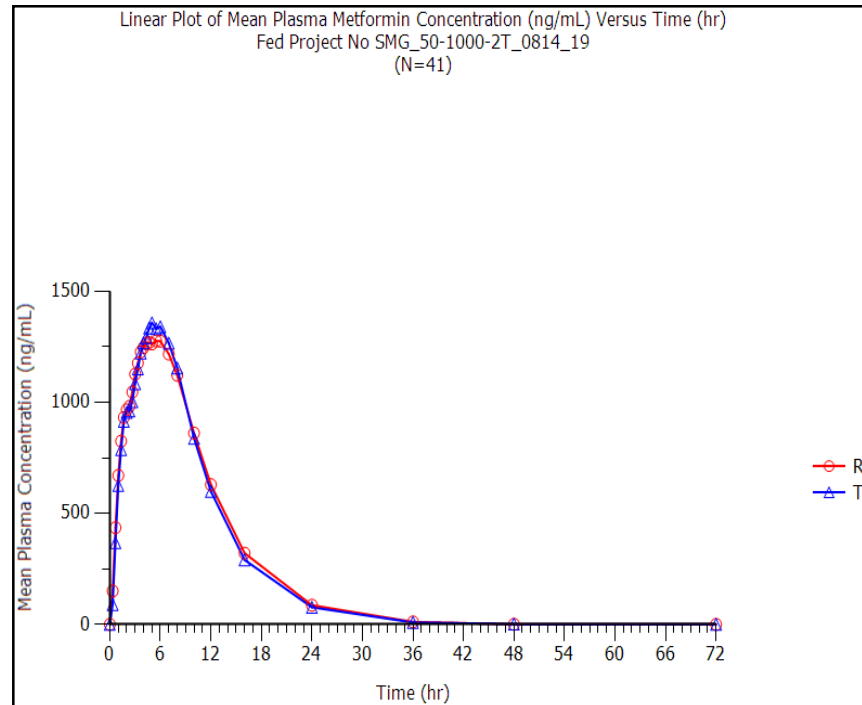
❑ Volunteers who met any one of the following criteria were not enrolled in the study.

- **Hypersensitivity:** history of hypersensitivity to Sitagliptin, Metformin, Glimepiride, or any related drugs.
- **Recurrent headaches and upper respiratory tract infection:** history of recurrent headaches and upper respiratory tract infection in the week before the study.
- **Drug-induced rash and/or pruritus:** history of drug-induced rash and/or pruritus.
- **Organ dysfunction or clinically significant abnormalities:** any evidence of organ dysfunction or clinically significant abnormalities.
- **Serious medical illnesses:** The subject should have no history of serious medical illnesses, including gastrointestinal, hepatic, renal, cardiovascular, pulmonary, neurological, or hematological diseases, as well as systemic diseases like diabetes, hypertension, sarcoidosis, rheumatic diseases, thyroid abnormalities, immunosuppression, glaucoma, or any serious potentially life-threatening illness.
- **Communication impairments:** any communication impairments that hinder the written informed consent process.
- **Smoking:** regular smoker (>10 cigarettes daily) or difficulty abstaining from smoking during the study period.
- **Drug dependence and alcohol intake:** history of drug dependence, excessive alcohol intake.
- **Medication use:** any medication, including oral contraceptives, within 30 days prior to the study.
- **Blood donation and investigational medicinal product:** The subject should not have donated blood or received an investigational medicinal product within 90 days prior to receiving the first dose of the study drug.
- **Alcohol and grapefruit:** consumed alcohol, grapefruit juice, or grapefruit supplements 48 hours prior to admission.
- **Pregnancy and breastfeeding:** If the subject is a female volunteer and has a positive pregnancy test prior to admission and currently breastfeeding, they should not participate in the study.

Bioavailability Study Results

Summary of Statistical Analysis – Metformin (T vs R) [N=41]

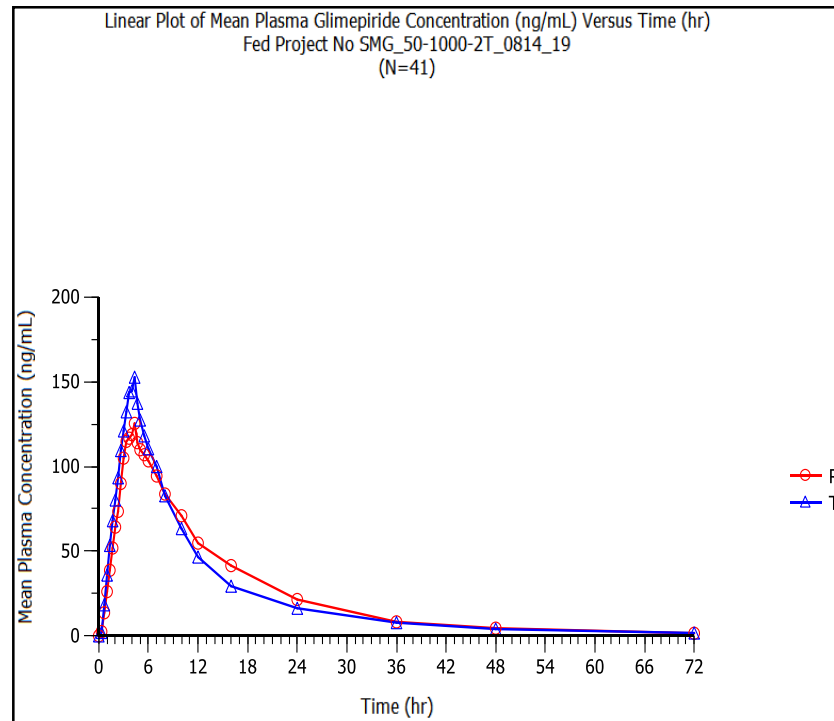
Parameter (units)	Geometric Least Square Mean		% Ratio (Tb/R)	90 %CI
	Test (Tb)	Reference (R)		
C _{max} (ng/mL)	1577.939	1500.847	105.02	97.80-112.76
AUC _{0-t} (ng.hr/mL)	15001.4520	15265.5145	98.23	94.26-102.37
AUC _{0-∞} (ng.hr/mL)	15313.8950	15560.0909	98.38	94.44-102.47
*T _{1/2} (h)	4.4316	4.4559		
#T _{max} (h)	5.000 (1.000-8.000)	5.000 (0.667-8.000)		
*Arithmetic mean presented; #Median (Min-Max) presented				



Metformin - Test product was equivalent to Reference product

Summary of Statistical Analysis – Glimepiride (T vs R) [N=41]

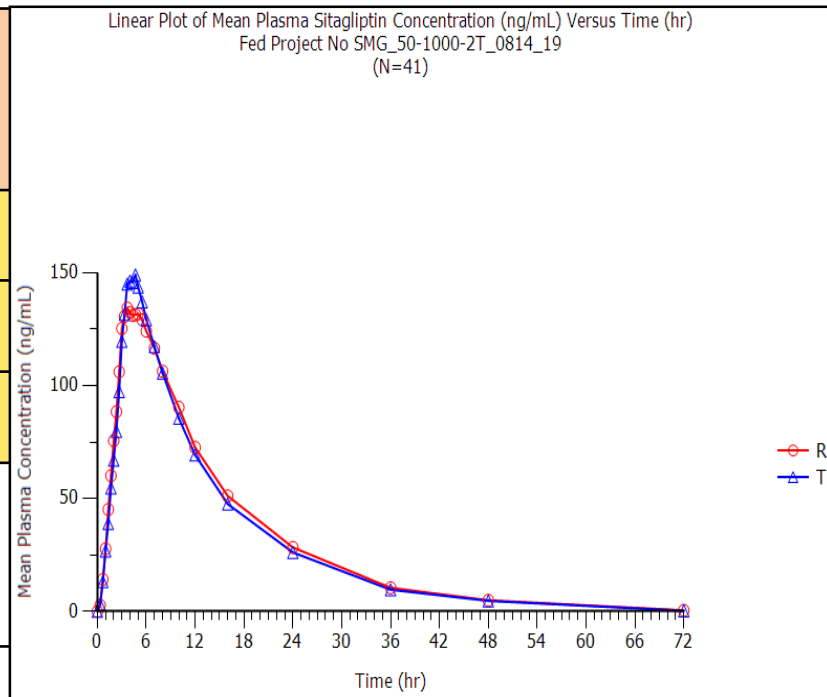
Parameter (units)	Geometric Least Square Mean		% Ratio (Tb/R)	90 %CI
	Test (Tb)	Reference (R)		
C _{max} (ng/mL)	169.591	1452.0437	113.12	103.15-124.06
AUC _{0-t} (ng.hr/mL)	1340.2635	1426.1117	94.00	89.57-98.65
AUC _{0-∞} (ng.hr/mL)	1369.8101	149.970	94.36	89.95-98.98
*T _{1/2} (h)	9.1959	8.1418		
#T _{max} (h)	4.333 (1.333-10.000)	4.333 (2.000-16.000)		
*Arithmetic mean presented; #Median (Min-Max) presented				



Glimepiride - Test product was equivalent to Reference product

Summary of Statistical Analysis – Sitagliptin (T vs R) [N=41]

Parameter (units)	Geometric Least Square Mean		% Ratio (Ta/R)	90 %CI
	Test (T)	Reference (R)		
C _{max} (ng/mL)	173.654	157.956	109.87	102.79-117.44
AUC _{0-t} (ng.hr/mL)	1973.3234	2031.3813	97.13	95.55-98.73
AUC _{0-∞} (ng.hr/mL)	2030.1054	2086.3015	97.29	95.70-98.91
*T _{1/2} (h)	9.6566	9.5208		
#T _{max} (h)	4.000 (1.000-10.017)	4.000 (1.333-10.000)		
* Arithmetic mean presented; #Median (Min-Max) presented				



Sitagliptin - Test product was equivalent to Reference product

- ❖ **Total of six (06) adverse events and One (01) medical event (before dosing) were reported in the study involving seven (07) subjects were reported in the study.**
- ❖ **No serious adverse event was reported during the conduct of this study.**

❑ **Adverse Events:**

- Subject no: 11 Heart Burn (Test T)
- Subject no: 35 Abnormal Jerky Movements (Test T)
- Subject no: 34 Decreased Platelets (Test T)
- Subject no: 48 Increased Urine WBC (Test T)
- Subject no: 01 Vomiting (Reference R)
- Subject no: 24 Increased Eosinophils (Reference R)

❑ **Medical Events:**

- Subject no: 23 Vomiting (Not Applicable)

- ❖ **Severity:** AEs were mild to moderate and non-serious in nature.
- ❖ **Relationship to study drug:** The AEs were judged unlikely and possibly related with drug administered.
- ❖ **Outcome:** AEs were recovered without sequelae.
- ❖ **No serious adverse event was reported in the study**

Our study results showed that, **the 90% confidence interval for the ratio of test (T) and reference (R) product averages (least squares means) for pharmacokinetic parameters (Cmax, AUC0-t and AUC0-Inf) were between 80% and 125% for the log-transformed data.**

- ❖ Based on these results, single oral dose of Sitagliptin Phosphate, Metformin Hydrochloride and Glimepiride 50 mg/ 1000 mg/ 2 mg tablets (Test Product) and Janumet® (Sitagliptin phosphate and Metformin Hydrochloride) tablets 50 mg/1000 mg [Reference Product (R1)] & Amaryl (Glimepiride) Tablets 2 mg [Reference Product (R2)] are **bioequivalent** in healthy adult human subjects under fed condition.
- ❖ The test (T) and reference (R) products were **well tolerated** by the study subjects. There was **no serious adverse event** reported in the study.

FDC Sitagliptin, Glimepiride and Metformin is expected to have significant implications in a real world setting by allowing decreasing dose of glimepiride, more reduction in HbA1c and reduce pill burden



Thank You